Task 4: High-Speed and Low-Speed Peripherals in SoC Designs

Objective
The objective of this task is to identify and describe commonly used high-speed and low-speed peripherals in System-on-Chip (SoC) designs. The peripherals are categorized according to their typical data rates, bus interface requirements, and applications.

High-Speed Peripherals
High-speed peripherals are designed to transfer large amounts of data or communicate with high-speed external devices. They generally require high-bandwidth interfaces and are commonly connected through high-performance buses such as AXI, AHB, PCIe, or dedicated high-speed interfaces.

Peripheral	Typical Data Rate	Bus / Interface	Applications
Ethernet	100 Mbps – 10 Gbps+	AXI/AHB + Ethernet MAC	Networking, internet connectivity, data communication
USB	12 Mbps – 10+ Gbps depending on version	USB + AXI/AHB	Storage devices, peripherals, data transfer
DDR Memory Controller	Several GB/s	AXI/AHB + DDR interface	Main memory and high-speed data storage
PCI Express (PCIe)	Several GB/s	PCIe + AXI	GPUs, NVMe storage, accelerators, high-speed expansion
SD/eMMC Controller	Tens to hundreds of MB/s	AXI/AHB + SD/eMMC	Embedded storage and data logging

1. Ethernet
Ethernet is a high-speed networking peripheral used for communication between an SoC and a local network or the internet. Depending on the implementation, Ethernet can support data rates ranging from 100 Mbps to multiple Gbps.
It commonly uses a dedicated Ethernet MAC connected to the SoC interconnect through AXI or AHB. Ethernet is widely used in routers, network devices, embedded systems, industrial controllers, and IoT gateways.

2. USB
USB provides communication between an SoC and external devices such as keyboards, storage

devices, cameras, and other peripherals. USB supports different generations with data rates ranging from low-speed USB 1.x modes to multi-Gbps USB 3.x and USB4 implementations.
A USB controller typically connects to the SoC through an internal high-performance bus such as AXI or AHB.

3. DDR Memory Controller

A DDR memory controller provides access to external dynamic RAM and is one of the most important high-bandwidth components in modern SoCs.
The controller interfaces with DDR3, DDR4, DDR5, LPDDR, or similar memory technologies. Internally, it commonly connects to the processor or other masters through AXI. It is used for program memory, buffers, operating systems, graphics, and large datasets.

4. PCI Express (PCIe)

PCIe is a high-speed serial interface used to connect an SoC to devices such as GPUs, NVMe storage, network adapters, and hardware accelerators.
PCIe provides high bandwidth and is commonly connected to an internal AXI-based interconnect through a PCIe controller or bridge.

5. SD/eMMC Controller

SD and eMMC controllers provide access to non-volatile storage devices. They are commonly used for boot storage, operating system storage, firmware, and data logging.The controller usually connects to the SoC through an AHB or AXI interface while communicating with the external storage using SD or eMMC protocols.

Low-Speed Peripherals

Low-speed peripherals generally transfer smaller amounts of data and do not require the bandwidth of high-speed interfaces. They are commonly connected through APB or similar low-power peripheral buses.

Peripheral	Typical Data Rate	Bus / Interface	Applications
UART	~9.6 kbps – several Mbps	APB/AHB + UART	Serial communication, debugging
SPI	Hundreds of kbps – tens of Mbps	APB/AHB + SPI	Sensors, displays, flash memory

I2C	100 kbps – 5 Mbps	APB + I2C	Sensors, EEPROMs, configuration devices
GPIO	Low / event dependent	APB	Digital input/output, control signals
PWM	Low-rate control signals	APB	Motor control, LEDs, power control

1. UART

UART is a simple serial communication peripheral commonly used for debugging, configuration, and communication with other devices. Typical baud rates range from 9.6 kbps to several Mbps. UART usually connects to the SoC through an APB or AHB interface.

2. SPI

SPI is a synchronous serial interface commonly used for communication with sensors, displays, ADCs, DACs, flash memory, and other peripherals. Depending on the hardware, SPI can support data rates from hundreds of kbps to tens of Mbps. The internal control registers are typically accessed through APB or AHB.

3. I2C

I2C is a two-wire serial communication protocol commonly used for low-speed communication with sensors, EEPROMs, RTCs, power-management ICs, and configuration devices. Common I2C modes include 100 kbps Standard Mode, 400 kbps Fast Mode, 1 Mbps Fast Mode Plus, and up to 3.4 Mbps High-Speed Mode. I2C controllers are typically connected to the processor through an APB interface.

4. GPIO

GPIO provides configurable digital input and output signals. It is one of the simplest peripherals in an SoC and is used for LEDs, buttons, interrupts, chip-select signals, and basic device control. GPIO does not have a continuous data rate like a serial communication interface. Its performance depends mainly on the SoC clock and GPIO implementation. GPIO control registers are commonly accessed through APB.

5. PWM

PWM generates periodic digital waveforms with configurable frequency and duty cycle. It is commonly used for motor control, LED brightness control, fan control, audio generation, and power-management applications. PWM typically has a low data/control bandwidth because only configuration registers need to be updated. It is commonly connected to the processor through APB.

Comparison

Category	High-Speed Peripherals	Low-Speed Peripherals
Main Purpose	Large-volume/high-bandwidth data transfer	Control, configuration, and small data transfers
Typical Bus	AXI/AHB	APB/AHB
Bandwidth	Mbps to multiple GB/s	Low kbps to tens of Mbps
Power Requirement	Generally higher	Generally lower
Examples	Ethernet, USB, DDR, PCIe, SD/eMMC	UART, SPI, I2C, GPIO, PWM

Conclusion

High-speed and low-speed peripherals serve different roles within an SoC. High-speed peripherals such as Ethernet, USB, DDR, PCIe, and SD/eMMC require high-bandwidth interfaces and are generally connected through performance-oriented buses such as AXI or AHB. Low-speed peripherals such as UART, SPI, I2C, GPIO, and PWM mainly perform control and configuration functions and are commonly connected through the low-power APB bus. Proper selection of the bus interface allows an SoC to achieve a balance between performance, power consumption, and hardware complexity.